5.5 Similarity Transformations

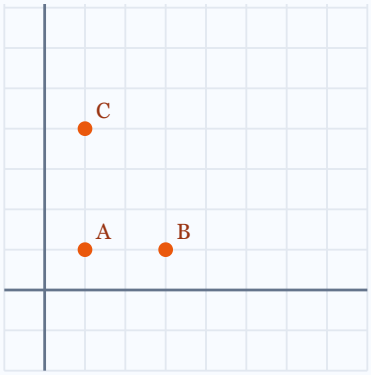
Two figures are similar exactly when a sequence of **rigid motions** (translations, rotations, reflections — which preserve size and shape) and a single **dilation** (which scales size by k but preserves shape) maps one onto the other. Rigid motions preserve every length and angle; a dilation preserves every angle but multiplies every length by $|k|$. Because of this, a similarity transformation always preserves angle measure, and scales all lengths by the same overall factor — even a compound sequence of several dilations has a single **net scale factor** equal to the product of the individual factors. When $k = 1$, a similarity transformation is simply a **congruence transformation**. Reflections are the one type of rigid motion that reverses orientation (clockwise ↔ counterclockwise), while translations and rotations preserve it. This worksheet practises identifying, composing, and reversing similarity transformations on the coordinate plane.

Examples

Example 1: Dilation followed by translation

$\triangle ABC$ has $A(1, 1)$, $B(3, 1)$, $C(1, 4)$. It is dilated by $k = 2$ about the origin, then translated by $(5, -1)$. Find the final image, and explain why it is similar to $\triangle ABC$.

Solution



Pre-image before dilation and translation.

Dilate each vertex by $k = 2$ about the origin: $A'(2, 2)$, $B'(6, 2)$, $C'(2, 8)$.

Translate by $(5, -1)$: $A''(7, 1)$, $B''(11, 1)$, $C''(7, 7)$.

The dilation scales every side by $k = 2$ while preserving angles; the translation is a rigid motion that changes neither size nor shape.

Conclusion: the final triangle has all angles congruent to $\triangle ABC$ and all sides exactly twice as long — it is similar to $\triangle ABC$ with scale factor 2.

Example 2: Describe the transformation sequence

$\triangle DEF$ has $D(0,0), E(2,0), F(0,3)$. It maps to $D'(4,6), E'(8,6), F'(4,12)$. Describe a similarity transformation that accomplishes this.

Solution

Find the scale factor using a side length: $DE = 2$; $D'E' = \sqrt{(8-4)^2 + (6-6)^2} = 4$. So $k = \frac{4}{2} = 2$.

Dilate each vertex by $k = 2$ about the origin first: $D \to (0,0), E \to (4,0), F \to (0,6)$.

Compare to the actual images: each is shifted by the same amount, $(4,6)$: $(0,0) + (4,6) = (4,6) = D'$ ✓, $(4,0) + (4,6) = (8,6) = E'$ ✓, $(0,6) + (4,6) = (4,12) = F'$ ✓.

Answer: dilate about the origin with $k = 2$, then translate by $(4,6)$.

Example 3: Scale factor from perimeter

A figure with perimeter 24 maps under a similarity transformation to a figure with perimeter 60. Find k. If a side of the original figure is 7, find the corresponding image side.

Solution

Perimeters of similar figures scale by the same factor as their sides: $k = \frac{60}{24} = \frac{5}{2}$.

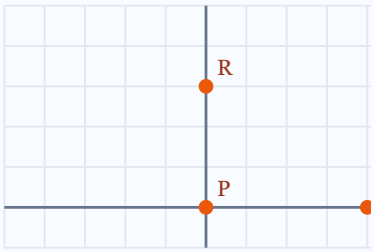
Apply k to the given side: image side $= 7 \times \frac{5}{2} = 17.5$.

Answer: $k = \frac{5}{2}$; corresponding image side $= 17.5$.

Example 4: Reflection reverses orientation

$\triangle PQR$ has $P(0,0)$, $Q(4,0)$, $R(0,3)$, listed counterclockwise. It is reflected over the y-axis. Find the image vertices and determine the new orientation.

Solution



Reflecting over the y-axis negates every x-coordinate.

Apply the reflection rule $(x, y) \mapsto (-x, y)$: $P'(0,0)$, $Q'(-4,0)$, $R'(0,3)$.
In the original, tracing $P \to Q \to R$ moves right then up-left — counterclockwise.
In the image, tracing $P' \to Q' \to R'$ moves left then up-right — the direction of travel has reversed.
Conclusion: the image is traversed clockwise — reflections always reverse orientation, unlike translations or rotations.

Example 5: The congruence special case

A triangle is rotated $90°$ about the origin, with no dilation applied. Explain why this is a similarity transformation with $k = 1$, and name the special case.

Solution

A rotation is a rigid motion — it preserves every length and angle exactly, with no scaling at all.
Since no length changes, the effective scale factor is $k = 1$.
Conclusion: a similarity transformation with $k = 1$ is simply a **congruence transformation** — congruence is the special case of similarity where nothing is resized.

Example 6: Find k from a pure dilation

Point $(3, 5)$ maps to $(9, 15)$ under a dilation about the origin (no other transformations). Find k, confirming with both coordinates.

Solution

Using the x-coordinates: $k = \frac{9}{3} = 3$.

Check with the y-coordinates: $k = \frac{15}{5} = 3$ ✓ — consistent.

Answer: $k = 3$.

Example 7: Composing two dilations

A figure is dilated about the origin by $k_1 = 2$, and the image is then dilated about the origin again by $k_2 = 3$. Find the net scale factor, and verify using the point $(2, 1)$.

Solution

The net scale factor of two dilations about the same center is the *product* of the individual factors: $k_{\text{net}} = k_1 \times k_2 = 2 \times 3 = 6$.

Apply step by step to $(2, 1)$: first dilation gives $(4, 2)$; second dilation gives $(12, 6)$.

Apply the net factor directly: $(2, 1) \times 6 = (12, 6)$ ✓ — matches.

Answer: net scale factor $= 6$.

Example 8: Scale factor from a rotation + dilation

A triangle has a side of length 6. After a rotation followed by a dilation, the corresponding image side measures 21. Find the scale factor k used in the dilation.

Solution

A rotation alone does not change any length, so all of the length change comes from the dilation.

$$k = \frac{\text{image length}}{\text{original length}} = \frac{21}{6} = 3.5.$$

Answer: $k = 3.5$.

Example 9: Logo scaling (word problem)

A company logo triangle has a base of 8 cm. A larger version for a billboard is created using a similarity transformation with $k = 4.5$. Find the billboard version's base length.

Solution

Apply the scale factor directly to the given length:

billboard base $= 8 \times 4.5 = 36$ cm.

Answer: 36 cm.

Example 10: Angle measure is preserved

$\triangle XYZ$ has $\angle X = 65°$. It is reflected over the x-axis, then dilated by $k = \frac{2}{5}$ about the origin. Find the measure of the corresponding angle in the final image, and explain why.

Solution

A reflection is a rigid motion, so it preserves angle measure exactly.

A dilation scales every length by k but leaves every angle measure unchanged (it only stretches distances, not the geometry of the angles).

Since neither transformation in the sequence changes angle measure, the final angle is unchanged.

Answer: $65°$ — similarity transformations always preserve angle measure, even when they scale lengths.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Scale factor

Corresponding sides 8 and 2. Scale factor from larger to smaller?

Answer: **Verification:**

Problem 2: Sequence

Name the two types of transformations that make up a similarity mapping.

Answer: **Verification:**

Problem 3: Side length

Under a similarity with $k = 3$, a side of length 5 becomes what length?

Answer: **Verification:**

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Problem 4: Angle

An angle measuring $110°$ maps under a similarity transformation to what measure?

Answer: **Verification:**

Problem 5: Congruence

A similarity transformation with $k = 1$ is also called what?

Answer: **Verification:**

Problem 6: Perimeter

Perimeter 18 maps to perimeter 45. Find k.

Answer: **Verification:**

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Problem 7: Dilation rule

Points $(2, 4)$ and $(4, 8)$ suggest what origin-centered scale factor?

Answer: **Verification:**

Problem 8: Orientation

Which type of rigid motion reverses orientation?

Answer: **Verification:**

Problem 9: Side

A side of 21 after a dilation with $k = \frac{1}{7}$ becomes what length?

Answer: **Verification:**

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Problem 10: Similar?

A figure is dilated by $k = 0.5$, then rotated. Is the image similar to the original?

Answer: **Verification:**

Problem 11: Scale factor

Corresponding sides 9 and 12. Find k from the first figure to the second.

Answer: **Verification:**

Problem 12: Fixed angles

Do similarity transformations always preserve angle measures?

Answer: **Verification:**

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Problem 13: Challenge — composing dilations

A figure is dilated about the origin by $k_1 = 1.5$, then dilated again about the origin by $k_2 = 4$. Find the net scale factor and the image of $(4, -2)$.

Answer: **Verification:**

Problem 14: Challenge — reflect then dilate

Point $(3, 7)$ is reflected over the x-axis, then dilated by $k = 3$ about the origin. Find the final image.

Answer: **Verification:**

Problem 15: Challenge — describe the sequence

$\triangle DEF$ with $D(0, 0), E(3, 0), F(0, 5)$ maps to $D'(2, 4), E'(8, 4), F'(2, 14)$. Find the scale factor and describe a dilation-then-translation sequence that works.

Answer: **Verification:**

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✓ Answer Key

1. $\frac{1}{4}$
2. Rigid motions and a dilation
3. 15
4. $110°$
5. A congruence transformation
6. $\frac{5}{2}$
7. $k = 2$
8. Reflection
9. 3
10. Yes
11. $\frac{4}{3}$
12. Yes
13. Net $k = 6$; image of $(4, -2)$ is $(24, -12)$
14. $(9, -21)$
15. $k = 2$; dilate about the origin with $k = 2$, then translate by $(2, 4)$

Great work — keep practising!